1.4.1 Conditioning of a linear system

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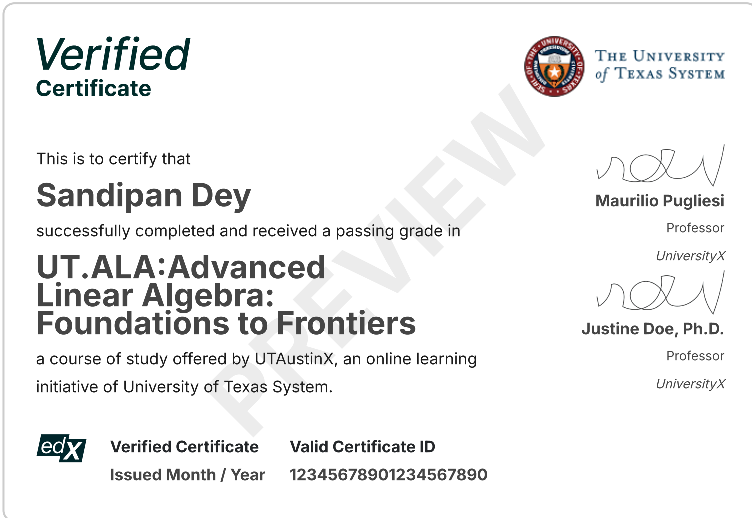

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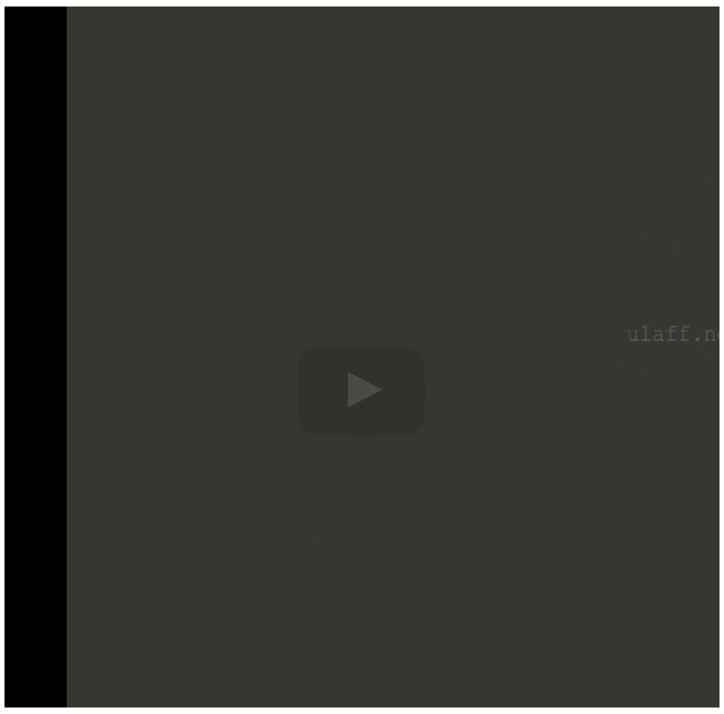

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2.0x

saying, "Let's bring the norm of x to the

left the norm of b to the right," And rewrite this as 1 over the norm of x is

less than or equal to the norm of A times 1 over the norm of b. That puts us

partially towards this right here. Alright.

Now, we need to somehow relate delta x and delta b, the norms And for that we're

actually going to rewrite A times delta x is equal to Delta b as

delta x is equal to A

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Read Unit 1.4.1 of the notes. [LINK]

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Homework 1.4.1.1

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Let $\|\cdot\|$ be a vector norm and corresponding induced matrix norm.

$\|I\| = 1.$

TRUE ✔ Answer: TRUE

For full solution see notes: [LINK]

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Answers are displayed within the problem

Homework 1.4.1.2

1/1 point (graded)
Let $\|\cdot\|$ be a vector norm and corresponding induced matrix norm, and A a nonsingular matrix.

$\kappa(A) = \|A\|\|A^{-1}\| \geq 1.$

TRUE ✔ Answer: TRUE

For full solution see notes: [LINK]

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